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Objective and Introduction

This report aims to do an avalanche risk assesment of the Kebnekaise mountain that is intuitive and easy to use for someone planning an touring excursion up the mountain. I decided to take on this topic for my final project because of a trip my friends and I are planning to do next year, where we will aim to climb the mountain and ski down it. Thus, the main objective of this project is, are there any safe paths up and (especially) down the mountain that will minimize the avalanche risk yet still be fun?

The risk assessment I conduct will be done using the points scaling system provided by the AVALUATOR slope evaluator 2.0, which is a risk determination card made by Avalanche Canada that can be taken up the mountain to determine avalanche risk on the code, coincidentally it is this exact card which I was taught to use for my avalanche training in Sweden.

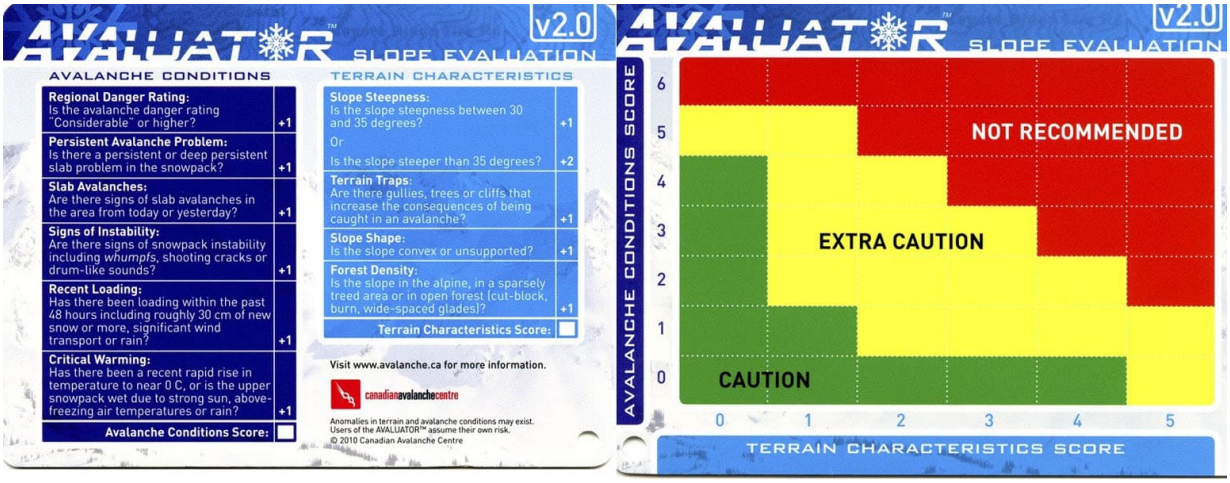


Figure 1: Avalauator Avalanche risk assesment card [1]

The analysis conducted in this report is with respect to the terrain features rather than avalanche/snow conditions. This is because I wanted to make the map a universal tool that can be used on any given day; and because the snow conditions are non constant variables, it would be unfitting to include based on the aim of my map. This being said, as part of the report I will include the redrawn map to depict every possible risk scenario based on the levels of avalanche/snow conditions risk which could exist, so as to give the reader a good visual interpretation of how the terrain risk scale and avalanche/snow condition risk scale relate to each other in any given scenario.

Data sources and their uses

As seen in the terrain characeristics in figure 1, the terrain factors that influence avalanche risk are slope steepness, terrain traps, slope shape, and forest density. For this report terrain traps such as trees and gullies are not considered because Lantmäteriet reports

none within the studied area, and it is for the same reason that forest density will not be considered either. This leaves the analysis with 3 main sources of danger, slope steepness, cliffs, and slope shape (convex or concave), thus making the maximum achievable terrain danger a 4. The listed data sources are given in table 1.

Table 1: Input Data

Data Type, Format	**Used For**	**Source**
Elevation data, 2 meters, Raster	Slope model, Cliffs model, curvature (convexivity) model, Contour lines	The Swedish Mapping, Cadastral and Land Registration Authority (Lantmäteriet), 2019 [3]
Mountain information, Vector	Registered walking trails	The Swedish Mapping, Cadastral and Land Registration Authority (Lantmäteriet), 2020 [3]

The elevation model was the main data source used for the analysis. Its derivative was computed to determine the slope, and its second derivative was used for the curvature. The curvature data was at first quite chaotic and granular as convexity greatly varied from pixel to pixel, so it was computed as an average for man neighborhoods spanning 25 meters to give a more useful final product.

The slope and the curvature were then reclassified based on the Avaluator card. For the slope this was done by creating a separate array that attributed a risk value of 0 for pixels with slope less than 30deg, a 1 for pixels between 30 and 35 deg, and a 2 for pixels with slope more than 35. The curvature was classified based on convexivity meaning that pixels with positive curvature got a 1 and with negative got a 0. This is because large amounts of snow have it easier to accumulate at the top of convex shaped slopes that on concave ones, which thereby causes a bigger avalanche when broken.

To the presence of cliffs was conveniently established via classification. Neighbourhoods of 3x3 pixels that had the properties of exceeding 75 degrees of slope as well as elevation difference greater than 5 meters were classified with a 1, and the rest with a 0.

These classifications were then summed together to create the terrain characteristic score. This score was then plotted. On the plot I have also included the registered walking paths, to make the map a more useful tool for mountaineers. Contour lines were also plotted on the map at 100m intervals, mainly to give a sense of the shape of the mountain independtly of the avalanche risk data.

Maps

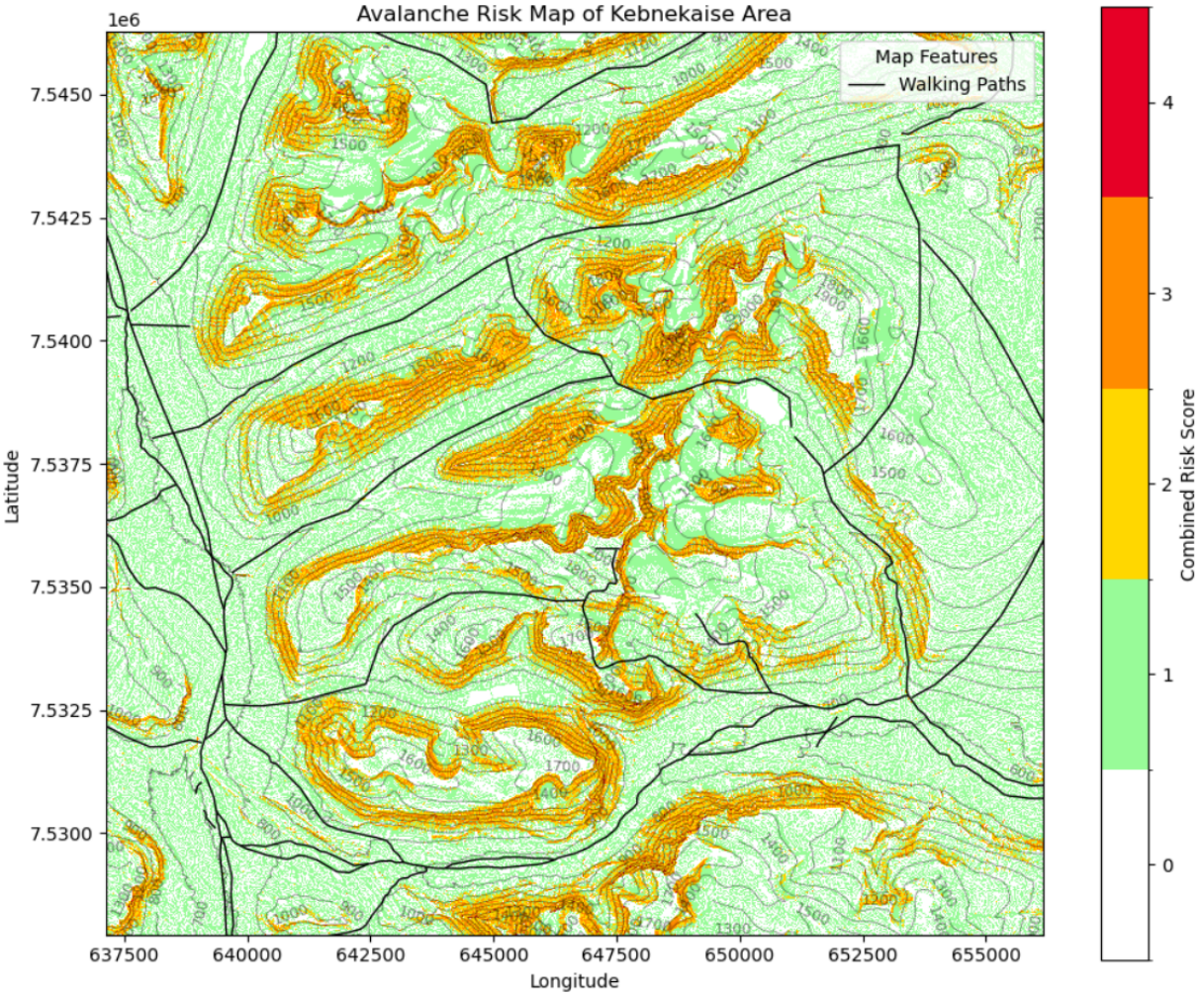


Figure 2: Avalanche terrain risk map of Kebnekaise

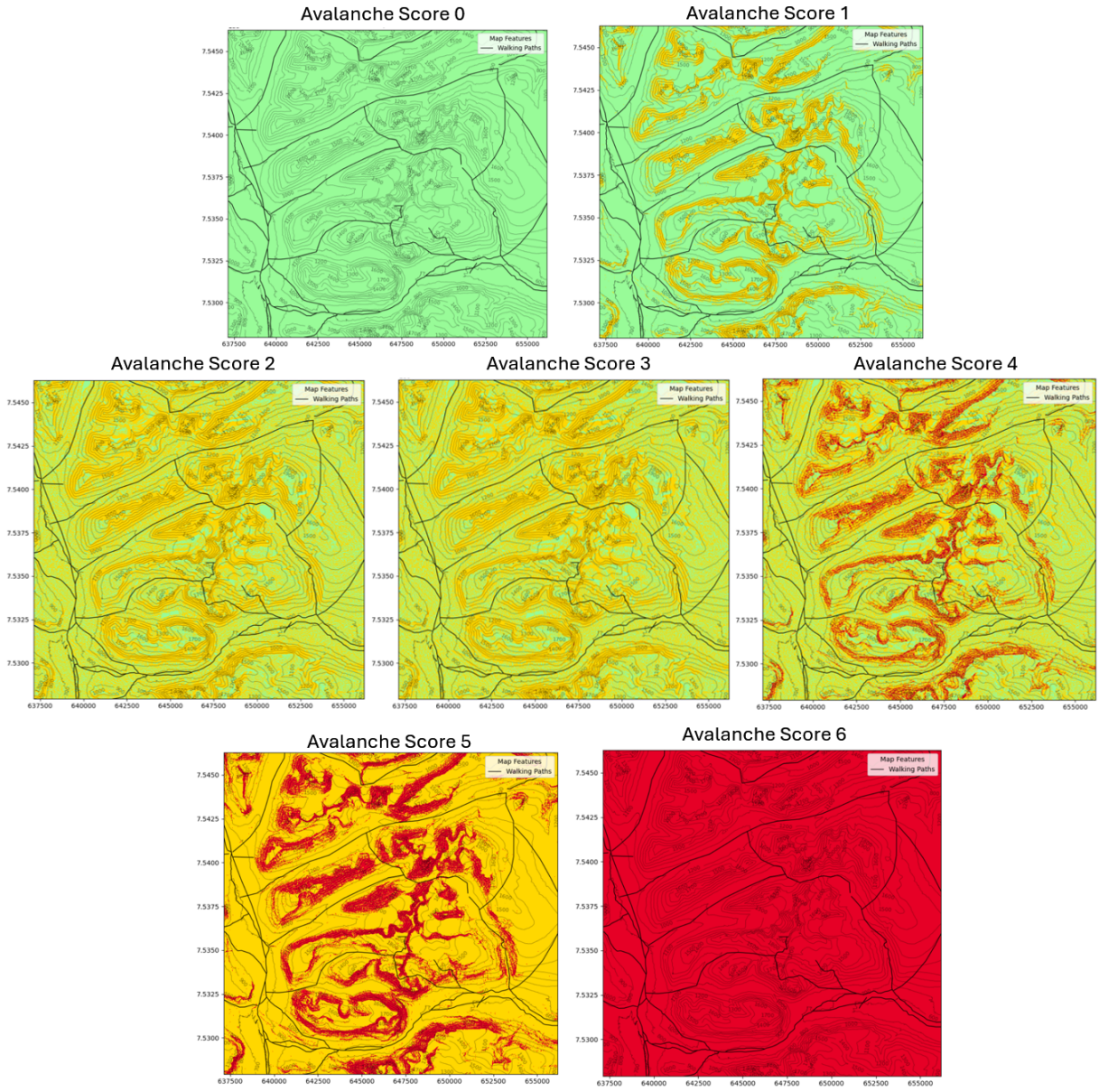


Figure 3: Avalanche terrain risk map of Kebnekaise with increasing avalanche snow risk from 0-6

References

[1] The avaluator. (n.d.). Retrieved January 25, 2025 https://avalanche.ca/pages/avaluator

[2] Lantmäteriet©. (n.d.). https://dike.slu.se/get/. Retrieved January 25, 2025, from https://dike.slu.se/get/